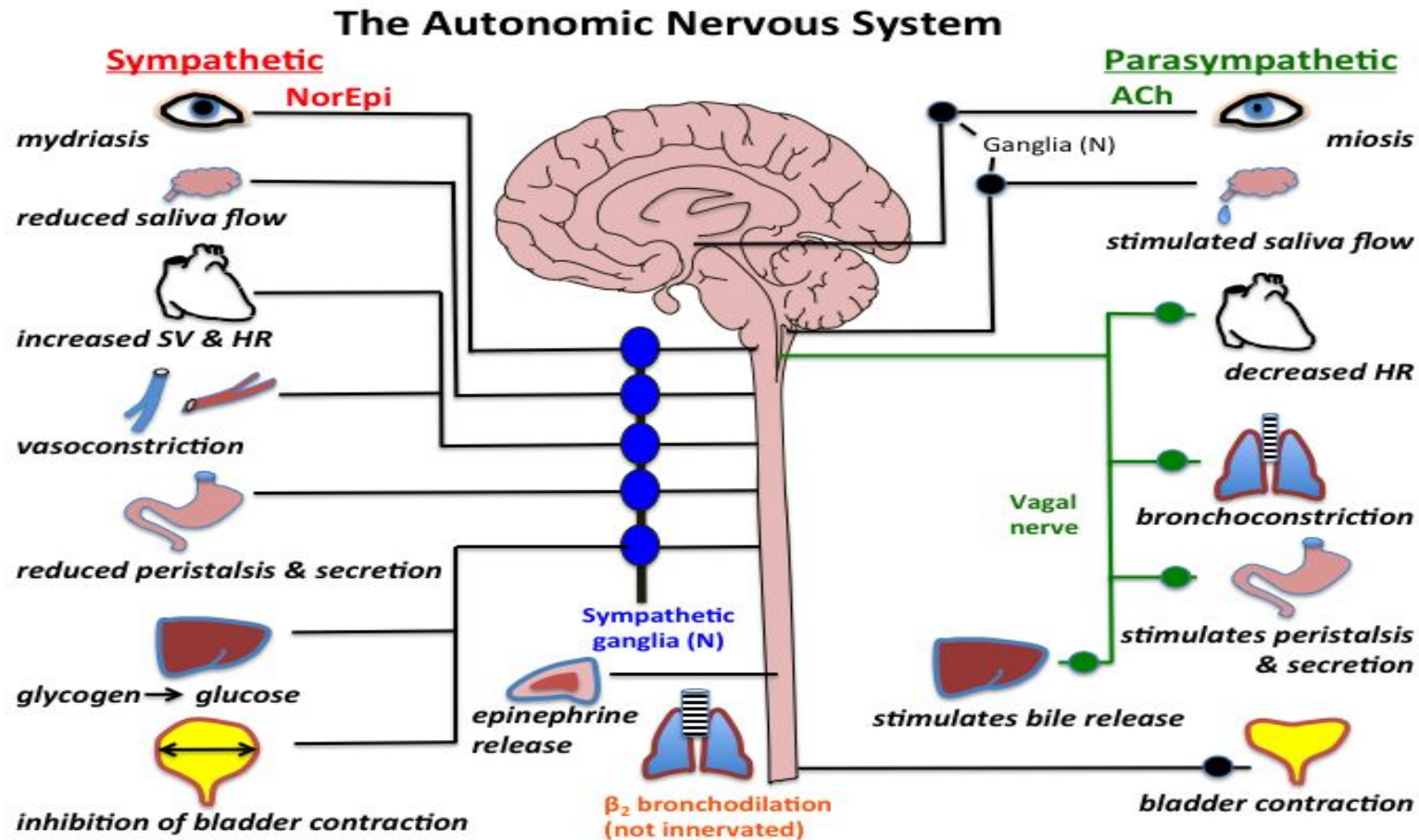
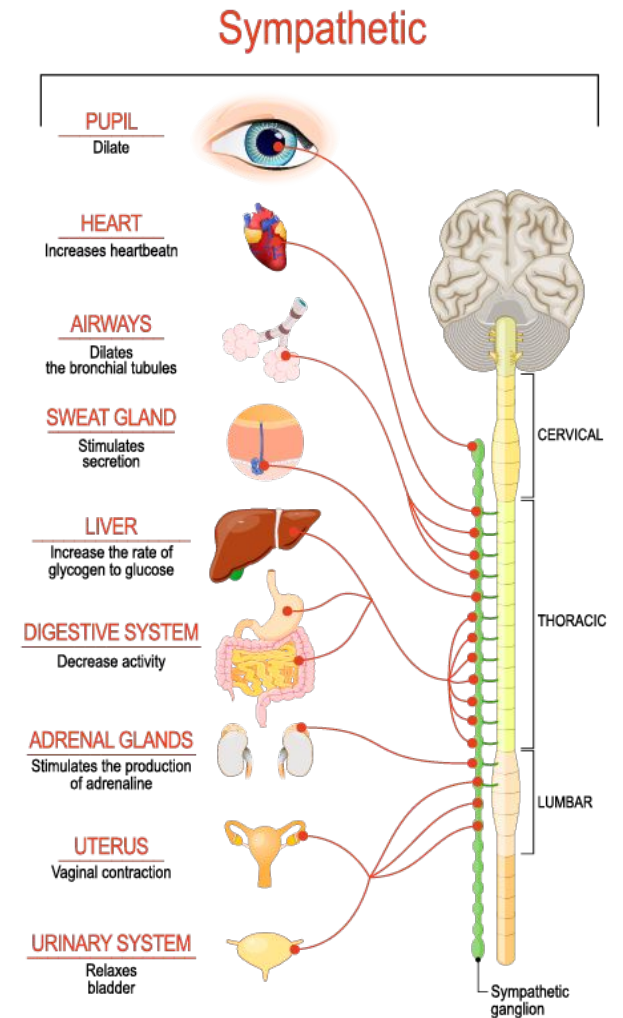
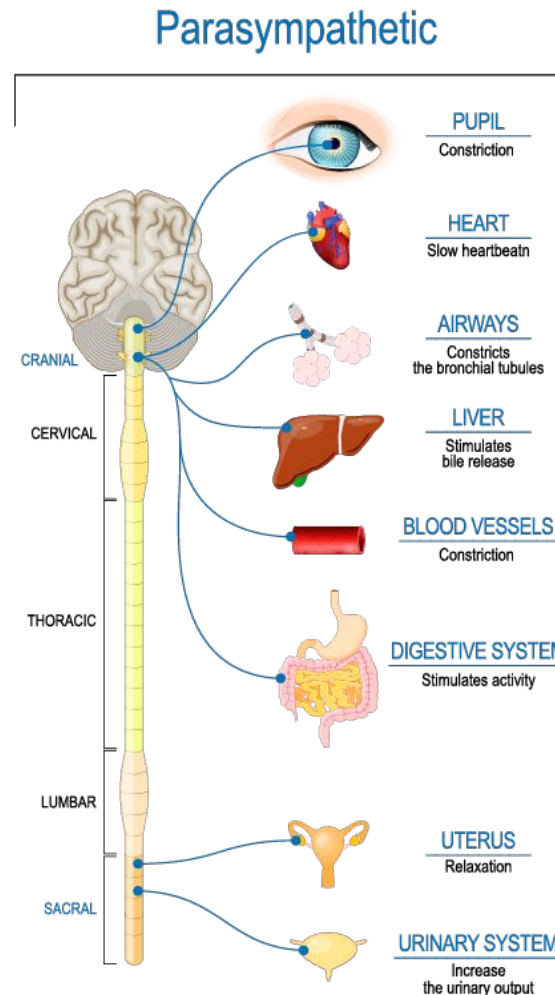


The Autonomic Nervous System



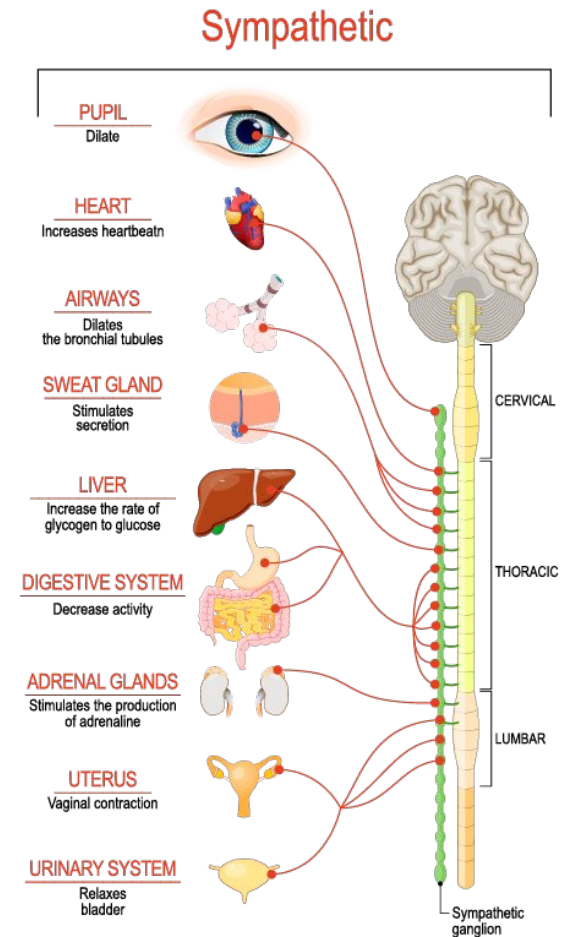
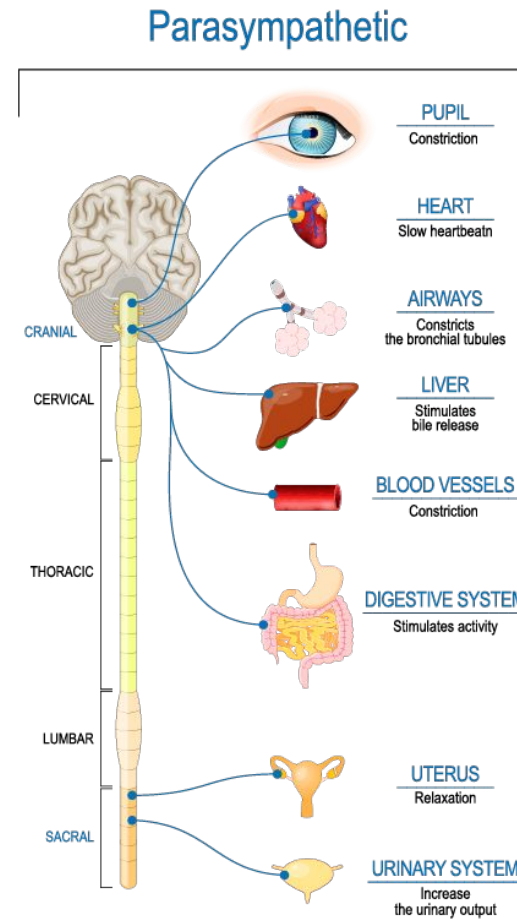
Definition:

- The part of the nervous system responsible for control of the bodily functions not consciously directed, such as breathing, the heartbeat, and digestive processes.
- It is a component of peripheral nervous system.
- The two divisions of autonomic nervous system are sympathetic division and parasympathetic division.



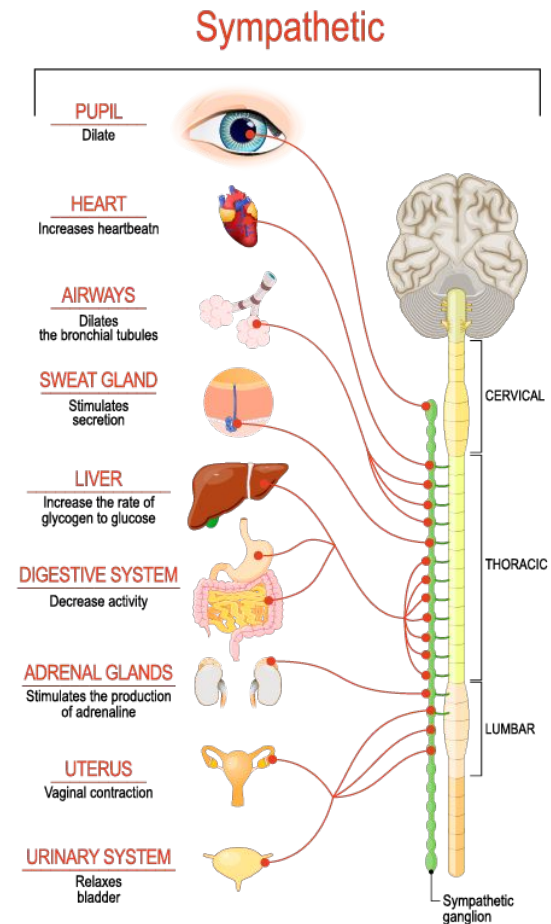
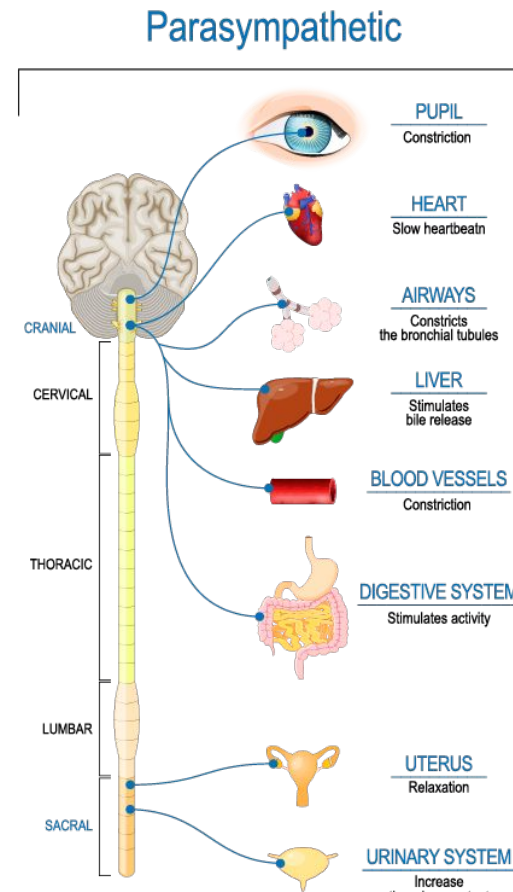
Parasympathetic nervous system:

- The parasympathetic nervous system, or craniosacral outflow, has its origin in neurons with cell bodies located in the brainstem nuclei of four cranial nerves—the oculomotor (cranial nerve III), the facial (cranial nerve VII), the glossopharyngeal (cranial nerve IX), and the vagus (cranial nerve X)—and in the second, third, and fourth segments of the sacral spinal cord.
- The preganglionic nerves arising from the brainstem as a part of the cranial nerves and travel through them to synapse with postganglionic neurons located in ganglia near or actually within the structures innervated.
- Preganglionic and postganglionic neurons release neurotransmitter acetylcholine (ACh).
- The parasympathetic nervous system is characterized by long preganglionic and very short postganglionic nerves.
- Governs 'rest & digest' response.



Sympathetic nervous system:

- Anatomically, the sympathetic preganglionic neurons, the cell bodies of which are located within the central nervous system, originate in the lateral horns of the 12 thoracic and the first 2 or 3 lumbar segments of the spinal cord.
- Preganglionic neurons of the spinal cord communicate with postganglionic sympathetic neurons via a series of sympathetic ganglia.
- Within the ganglia, spinal cord (preganglionic) sympathetic neurons join postganglionic sympathetic neurons through chemical synapses.
- Preganglionic neurons of sympathetic nervous system release acetylcholine whereas postganglionic neurons utilize epinephrine/norepinephrine.
- The sympathetic nervous system is characterized by short preganglionic and long postganglionic nerves.
- Governs 'fight or flight' response.



Differences between: sympathetic & parasympathetic nervous system

- | | |
|---|---|
| 2. Sympathetic nervous system works through neurotransmitter, adrenaline. | 2. Parasympathetic nervous system works through release of acetylcholine. |
| 3. Sympathetic nervous system enhances all the involuntary functions. | 3. Parasympathetic nervous system retards all the involuntary functions. |
| 4. It brings about fight, fright and flight responses. | 4. It brings about relaxation, comfort, pleasure, etc. |
| 5. The pre-ganglionic nerve fibres are short and the post-ganglionic nerve fibres are long in sympathetic nervous system. | 5. The pre-ganglionic nerve fibres are long and the post-ganglionic nerve fibres are short in parasympathetic nervous system. |

Functions of ANS

Sympathetic

- ▶ Heart
 - ▶ ↑ heart rate
 - ▶ ↑ force of contraction
- ▶ Blood vessels
 - ▶ *Constriction*
- ▶ Lungs
 - ▶ Bronchodilation
- ▶ GIT
 - ▶ ↓ motility
 - ▶ Sphincter contraction
 - ▶ Decreased secretions

Parasympathetic

- ▶ Heart
 - ▶ ↓ heart rate
 - ▶ ↓ force of contraction
- ▶ Blood vessels
 - ▶ No effect
- ▶ Lungs
 - ▶ Bronchoconstriction
- ▶ GIT
 - ▶ ↑ motility
 - ▶ Sphincter relaxation
 - ▶ Increased secretions



**Thank
you**